50-mavzu

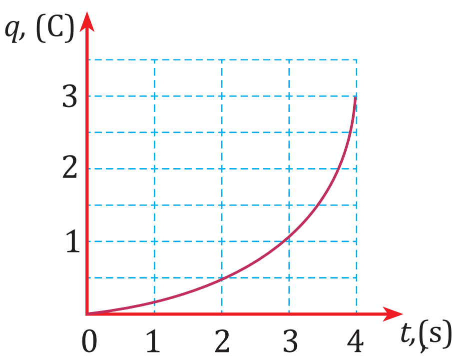
MASALALAR YECHISH

1 Manbadan o‘tkazgich uchlariga 3V kuchlanish berildi. Agar shu o‘tkazgichda 0,5 soat davomida 120 mA tok o‘tib turgan bo‘lsa, tok manbai zaryadni ko‘chirishda qanday ish bajargan?

Berilgan:	Formula	Hisoblash
$U = 3$ V $t = 0{,}5$ h $= 1800$ s $I = 120$ mA $= 0{,}12$ A **Topish kerak:** $A = ?$	$I = \dfrac{q}{t}$; $q = I \cdot t$; $A = q \cdot U = I \cdot t \cdot U$ $[A] = [q \cdot U] = \mathrm{C} \cdot \mathrm{V} = \mathrm{J}$	$A = 0{,}12 \cdot 1800 \cdot 3$ J $= 648$ J **Javob:** $A = 648$ J.

2 Tok manbaiga ulangan o‘tkazgichdan 3,2 A tok o‘tib turibdi. 30 minut davomida shu o‘tkazgich ko‘ndalang kesim yuzasidan o‘tgan jami elektronlarning massasini aniqlang.

Berilgan:	Formula	Hisoblash
$I = 3{,}2$ A $t = 1800$ s $\lvert e \rvert = 1{,}6 \cdot 10^{-19}$ C $m_0 = 9{,}1 \cdot 10^{-31}$ kg **Topish kerak:** $m = ?$	$I = \dfrac{q}{t}$; $q = N \cdot e$; $N = \dfrac{q}{e} = \dfrac{I \cdot t}{e}$; $m = N \cdot m_0 = \dfrac{I \cdot t}{e} \cdot m_0$ $[m] = \dfrac{\mathrm{A \cdot s}}{\mathrm{C}} \cdot \mathrm{kg} = \dfrac{\mathrm{C}}{\mathrm{C}} \cdot \mathrm{kg} = \mathrm{kg}$	$m = \dfrac{3{,}2 \cdot 1800}{1{,}6 \cdot 10^{-19}} \cdot 9{,}1 \cdot 10^{-31}\,\mathrm{kg} \approx 3{,}3 \cdot 10^{-8}\,\mathrm{kg}$ **Javob:** $m \approx 3{,}3 \cdot 10^{-8}$ kg.



3 Grafikda o‘tkazgichning ko‘ndalang kesim yuzidan o‘tgan zaryad miqdorining vaqtga bog‘liq grafigi keltirilgan. Vaqtning $t = 4$ s bo‘lgan paytida o‘tkazgichdagi tok kuchini aniqlang.

Berilgan:	Formula	Hisoblash
$t = 4$ s $q = 3$ C **Topish kerak:** $I = ?$	$I = \dfrac{q}{t}$; $[I] = \dfrac{\mathrm{C}}{\mathrm{s}} = \mathrm{A}$	$I = \dfrac{3}{4}\,\mathrm{A} = 0{,}75\,\mathrm{A}$ **Javob:** $I = 0{,}75$ A.

30-mashq

1 Elektr lampochkadan 0,8 A tok o‘tmoqda. Uning spirali ko‘ndalang kesim yuzasidan 10 minutda o‘tgan elektronlarning massasini aniqlang.

2 Manbaga ulangan iste’molchidan 20 mA tok o‘tib turibdi. Tok manbai 2 soat davomida zaryadni ko‘chirishda 720 J ish bajargan bo‘lsa, iste’molchi uchlariga qanday kuchlanish berilgan?

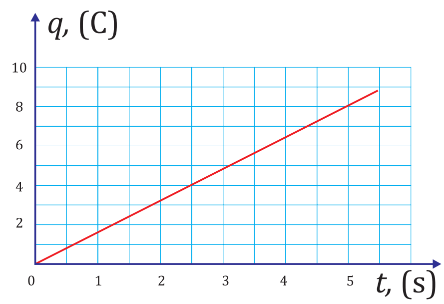
3 Elektr zanjirdagi lampochkadan 0,4 A tok o‘tmoqda. Lampochka spirali orqali uch minutda qancha zaryad va nechta elektron o‘tishini hisoblang.

4 12 V kuchlanishli akkumulyator avtomobilni yurgizishda generatorga 50 A tok bermoqda. Agar avtomobil dvigateli 2 s o‘tgach o‘t olsa, akkumulyator qanday ish bajargan?

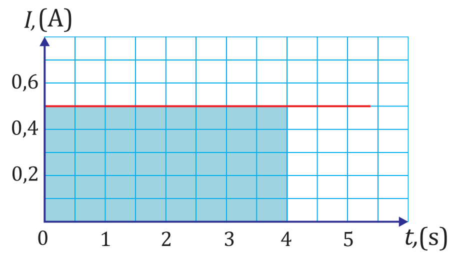
5 Elektr zanjiriga ulangan elektr lampochkadan ma’lum vaqt davomida 25 C zaryad o‘tib, tok manbai 100 J ish bajardi. Lampochka spiraliga qanday kuchlanish berilgan?

6 4.50-rasmda o‘tkazgich orqali o‘tgan zaryad miqdorining vaqtga bog‘liqlik grafigi keltirilgan. Grafik asosida o‘tkazgichdagi tok kuchini aniqlang.

7 4.51-rasmda tok kuchining vaqt bo‘yicha grafigi keltirilgan. Grafik asosida 4 s davomida o‘tkazgichdan o‘tgan zaryad miqdorini aniqlang.



4.50-rasm



4.51-rasm